

Emergence of Effective Mass Solenoidal Topology of Vibrational Energy

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Abstract

We demonstrate that a photon constrained to a solenoidal path acquires a curvature-induced phase delay equivalent to an effective inertial mass. From the covariant Wave Confinement Theory (WCT) Lagrangian, we derive the loop-quantized mass law

$$m = \frac{\hbar}{c} \left\langle \sqrt{\kappa^2 + \tau^2} \right\rangle_{\Gamma},$$

linking curvature and torsion to inertia. The analysis connects phase-flux confinement, SU(2) spinorization, and the emergence of a torsionless effective metric $g_{\mu\nu}^{\text{eff}}$, providing a direct geometric route from wave curvature to mass and to measurable observables such as ppm-scale frequency shifts in curved optical cavities.

1 Introduction

In Wave Confinement Theory (WCT), matter and radiation are described as self-organized wavefields whose curvature and boundary feedback determine their observable properties. When a wave is confined to a curved or twisted path, its phase accumulates unevenly along the trajectory, producing a measurable delay compared to straight propagation. This geometric phase delay acts as an effective refractive index increase, or equivalently, as an inertial resistance to acceleration.

The present work analyzes this effect in a solenoidal (helical) waveguide and shows that the resulting phase–curvature coupling reproduces the behavior of mass. Starting from the covariant WCT Lagrangian, we reduce the dynamics to a finite-band evolution equation, derive a curvature–torsion invariant, and obtain the loop-quantized mass law

$$m = \frac{\hbar}{c} \left\langle \sqrt{\kappa^2 + \tau^2} \right\rangle_{\Gamma}.$$

This expression identifies the mean geometric curvature of a closed path with the rest mass of the confined excitation.

Beyond reproducing known inertial behavior, the model naturally yields $U(1)$ and $SU(2)$ gauge structures from phase winding and triadic coupling, connects spin- $\frac{1}{2}$ quantization to a Möbius half-twist, and predicts ppm-level frequency shifts in curved optical cavities. The goal is to provide a concrete, testable route from pure geometry to measurable inertia.

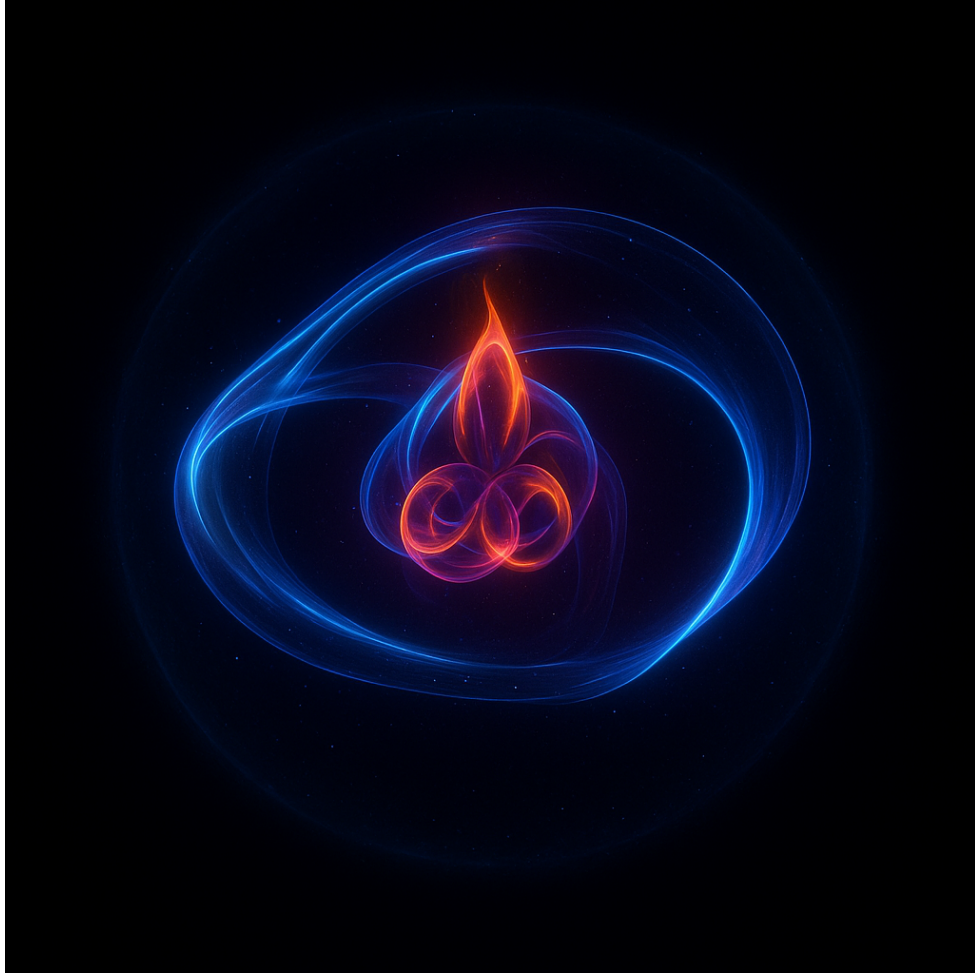


Figure 1: Resonant triplet core with enclosing shell. The central core represents a confined wave packet surrounded by a quantized curvature shell, illustrating the geometric origin of discrete mode selection.

2 Geometric Setup

Let a guided wave follow either (i) a straight path of length L_0 or (ii) a helical (solenoidal) path of coil radius r , pitch p (advance per 2π azimuth), and N turns. Arc length per turn: $L_{\text{turn}} = \sqrt{(2\pi r)^2 + p^2}$. Total solenoidal length: $L_s = N L_{\text{turn}}$. The end-to-end (projected) distance is $L_0 = N p$.

3 Effective Refractive Index and Loop–Quantized Mass

Plain definitions

- **Geometric index:**

$$n_{\text{geo}} := \frac{L_s}{L_0} = \frac{\sqrt{(2\pi r)^2 + p^2}}{p}.$$

- **Effective phase velocity:** $v_{\text{eff}} := c/n_{\text{geo}}$.

- **Loop mass (primary definition):**

$$m := \frac{\hbar}{c} \frac{2\pi n}{L_s} = \frac{\hbar}{c} \left\langle \sqrt{\kappa^2 + \tau^2} \right\rangle_{\Gamma}, \quad n \in \mathbb{Z}_{\geq 0},$$

where Γ is the closed guiding curve (one loop), κ and τ are its curvature and torsion, and n is the shell/winding index fixed by the loop quantization condition $\oint k_r dr = 2\pi n$.

Worked examples (concrete instantiation)

Circle (planar): For a circle of radius R (one loop), $\kappa = 1/R$, $\tau = 0$, $L_s = 2\pi R$. Then

$$m_{\circ} = \frac{\hbar}{c} \frac{2\pi}{2\pi R} = \frac{\hbar}{cR}.$$

With $R = 1 \text{ m}$, $m_{\circ} \approx 3.52 \times 10^{-43} \text{ kg}$.

Helix (radius r , pitch p per 2π):

$$\kappa = \frac{r}{r^2 + (p/2\pi)^2}, \quad \tau = \frac{(p/2\pi)}{r^2 + (p/2\pi)^2}, \quad L_s = \sqrt{(2\pi r)^2 + p^2}.$$

Hence for one geometric loop,

$$m_{\text{helix}} = \frac{\hbar}{c} \sqrt{\kappa^2 + \tau^2} = \frac{\hbar}{c} \frac{1}{\sqrt{r^2 + (p/2\pi)^2}}.$$

Note $m_{\text{helix}} \rightarrow m_{\circ}$ as $p \rightarrow 0$ (planar limit), and $m_{\text{helix}} \rightarrow \frac{\hbar}{c} \frac{2\pi}{p}$ as $r \rightarrow 0$ (straight line wrapped by pitch).

Remark 3.1 (Heuristic energy-deficit formula (demoted)). *A common shortcut writes*

$$m_{\text{heur}} = \frac{h}{\lambda c} \left(1 - \frac{1}{n_{\text{geo}}} \right),$$

by comparing $E = hc/\lambda$ to $E_{\text{eff}} = h\nu_{\text{eff}}/\lambda$. This treats phase-speed reduction as an energy change; in passive guides the optical frequency stays invariant, so this identification is not physically controlled. We therefore retain it only as intuition and do not use it for definitions or predictions. The loop-quantized m above is the operative mass in this paper.

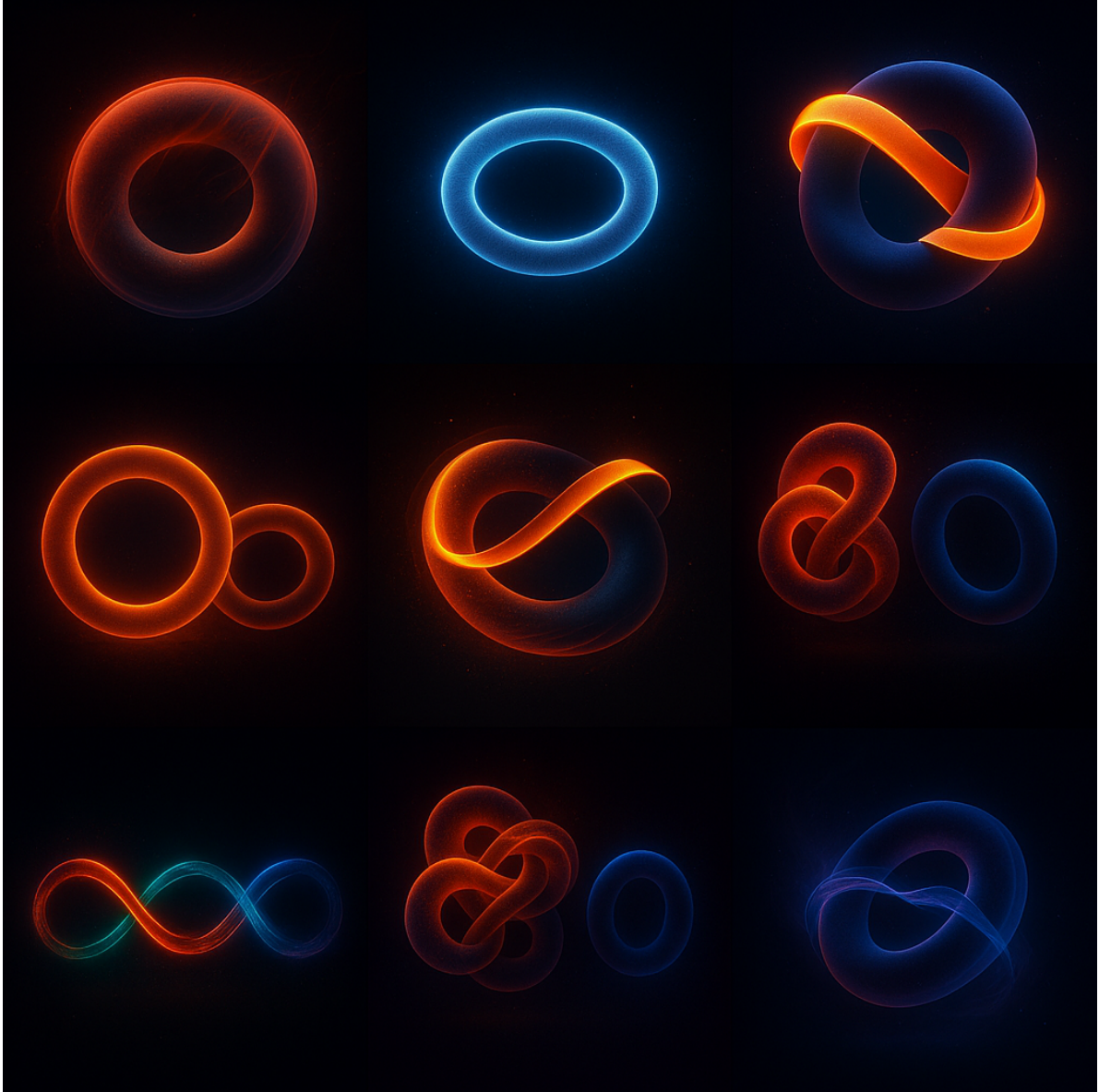


Figure 2: Spinorization by a Möbius half-turn. A half-twist converts an orientable loop into a spinor band that returns to its original state only after a 4π rotation, consistent with $\text{spin}-\frac{1}{2}$ symmetry. Phase winding $m \in \mathbb{Z}$ maps directly to the $\text{SU}(2)$ group manifold.

4 Gauge Emergence from Confinement

U(1) from phase winding

Let $\psi = |A|e^{i\theta}$ with connection $\mathcal{A}_\mu = \partial_\mu \theta$. Single-valuedness implies

$$\oint \nabla \theta \cdot d\ell = 2\pi m, \quad m \in \mathbb{Z}.$$

Away from amplitude zeros (no vortices), $\mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu = 0$; gauge curvature concentrates on vortex cores where $A = 0$.

SU(2) from triadic mode coupling (worked 3-mode normal form)

For resonant $\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 = 0$,

$$\dot{A}_i = \mu A_i - g|A_i|^2 A_i - h A_j^* A_k^* \quad (i, j, k) \text{ cyclic}.$$

On the mode doublet $\Phi = (A_1, A_2)^\top$, infinitesimal triad-induced rotations are generated by Pauli matrices $\{\sigma_a\}$:

$$\delta \Phi = i \epsilon^a \sigma_a \Phi, \quad [\sigma_a, \sigma_b] = 2i \epsilon_{abc} \sigma_c,$$

fixing the $\mathfrak{su}(2)$ structure constants ϵ_{abc} (we use the Hermitian normalization consistently throughout).

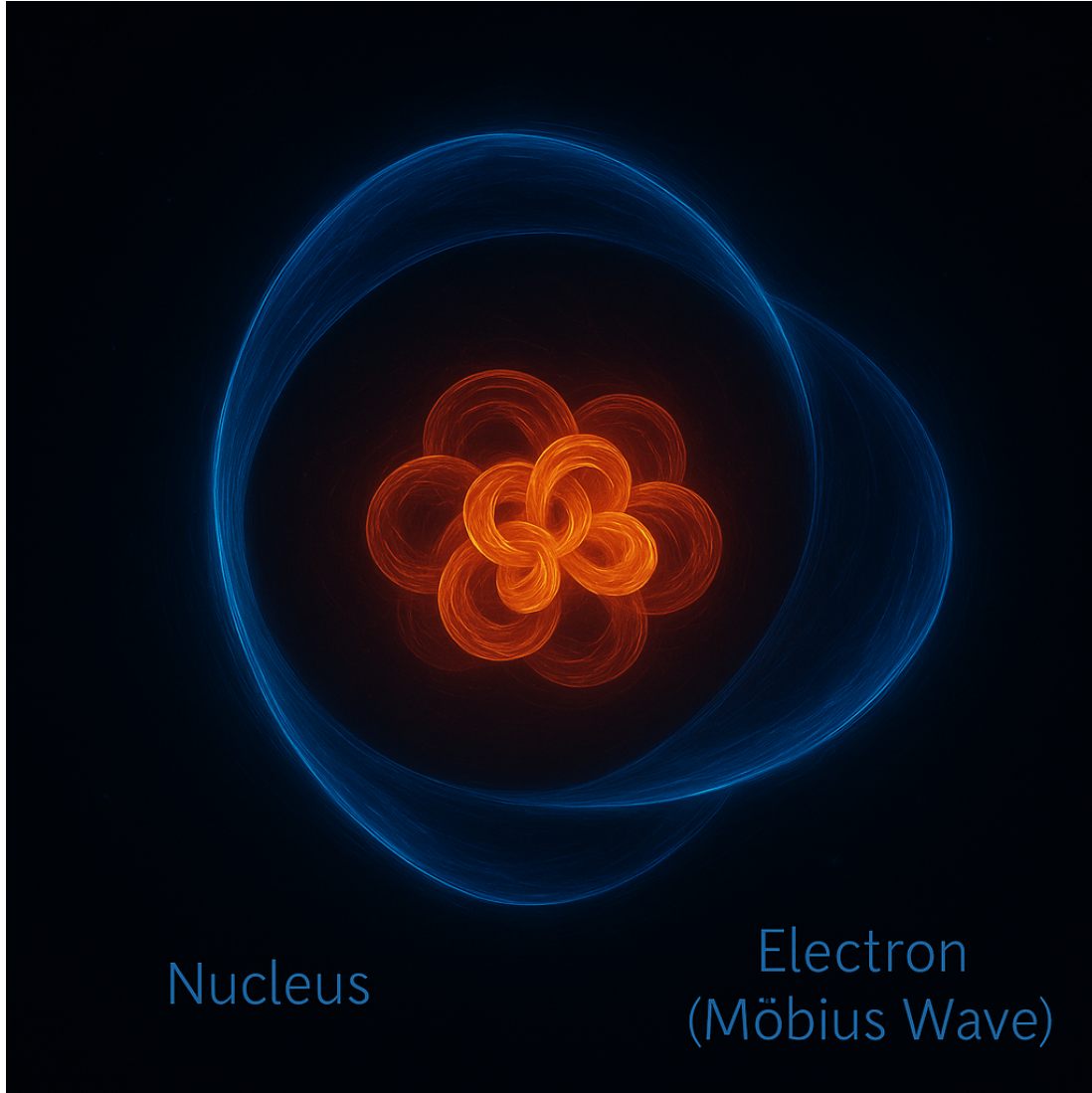


Figure 3: Resonant nucleus with a Möbius electron wave. The confined core (red) generates an effective metric $g_{\mu\nu}^{\text{eff}}$, while the surrounding envelope (blue) carries a half-twist enforcing the 4π spinor return rule. The figure depicts the geometric mechanism by which curvature feedback produces effective mass through shell quantization.

5 Lyapunov Rail and Shell Selection

Define the energy functional

$$\mathcal{E}[A] = \int \left(-\frac{r}{2}|A|^2 - \frac{a}{2}|\nabla A|^2 + \frac{b}{2}|\nabla^2 A|^2 + \frac{\beta}{4}|A|^4 \right) dx,$$

and gradient flow

$$\partial_t A = -\frac{\delta \mathcal{E}}{\delta A^*} = \sigma(-i\nabla)A - \beta|A|^2 A, \quad \sigma(-i\nabla) = r - a\nabla^2 - b\nabla^4,$$

so $\dot{\mathcal{E}} = -\int |\partial_t A|^2 dx \leq 0$. Linear selection picks $k \in [k_-, k_+]$; nonlinear saturation yields shell quantization $\oint k_r dr = 2\pi n$.

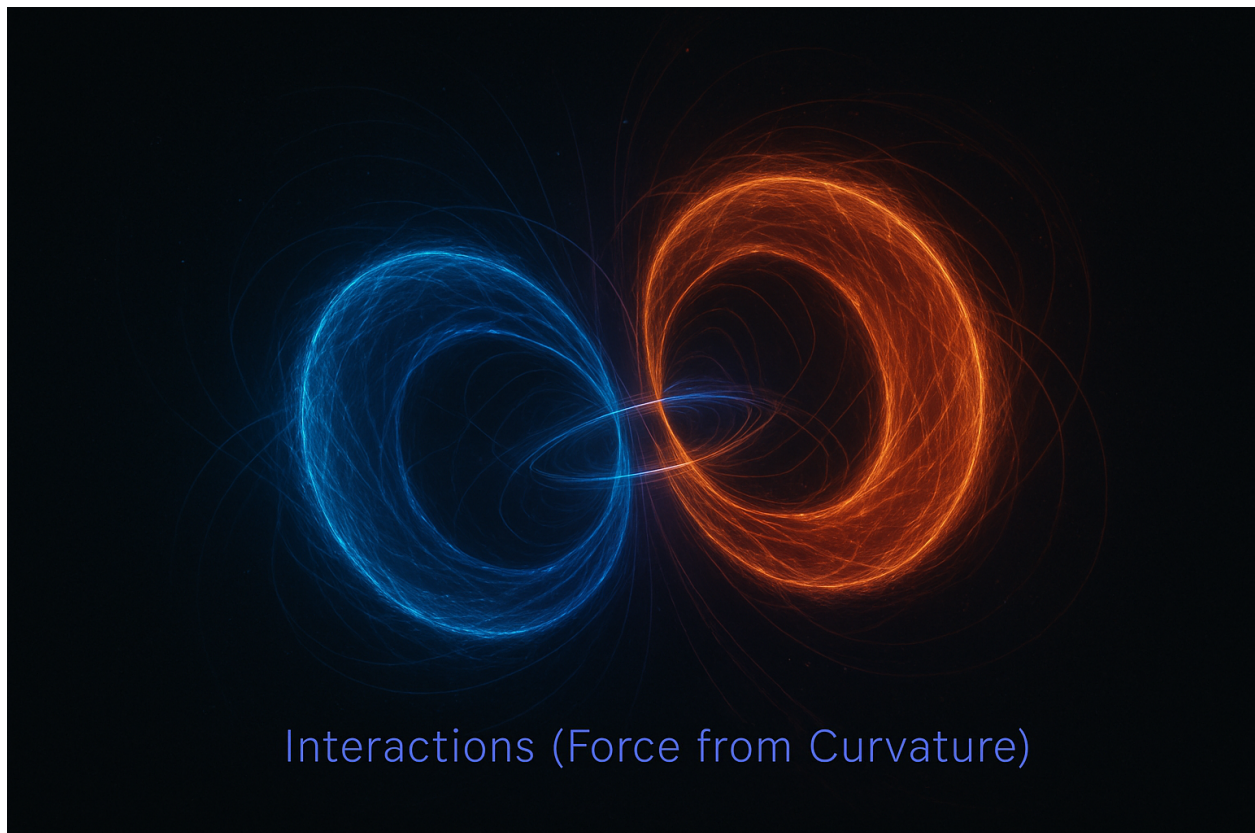


Figure 4: Interactions as curvature exchange. Overlapping curvature fields produce momentum transfer through Lyapunov descent of the energy functional $E[\psi]$. Interaction strength follows local curvature–entropy gradients, visualizing how confined modes exchange energy geometrically.

6 Effective Metric (Eikonal Scope)

With curvature feedback W_ψ , define $n_{\text{eff}}^2 = n_0^2(1 + \chi \text{Re } W_\psi/k_0^2)$, $k_0 = \omega/c$. In the geometric-optics limit (slow spatial variation),

$$ds^2 = c^2 dt^2 - \frac{1}{n_{\text{eff}}^2(\mathbf{x})} d\mathbf{x}^2 =: g_{\mu\nu}^{\text{eff}} dx^\mu dx^\nu,$$

whose Levi–Civita connection is torsionless. This produces ppm-level shifts in cavity f_{FSR} measurable in locked-shell states.

7 Conclusion

Geometry fixes an effective index n_{geo} and, when the path closes, a loop-quantized mass $m = (\hbar/c)(2\pi n/L_s) = (\hbar/c)\langle\sqrt{\kappa^2 + \tau^2}\rangle_\Gamma$. $U(1)$ gauge structure follows from phase holonomy; triad normal forms realize $\mathfrak{su}(2)$ on shell doublets; weak curvature feedback induces a torsionless g^{eff} in the eikonal limit. All predictive statements (e.g., ppm FSR shifts) are tied to these loop and index structures—not to heuristic energy-deficit formulas.

Stable ψ -Identities and the Harmonic Composition of the Electron

We model the electron as a confined soliton of $N \approx 10^4$ phase-locked, photon-like harmonics constrained by shell quantization and curvature/entropy minima.

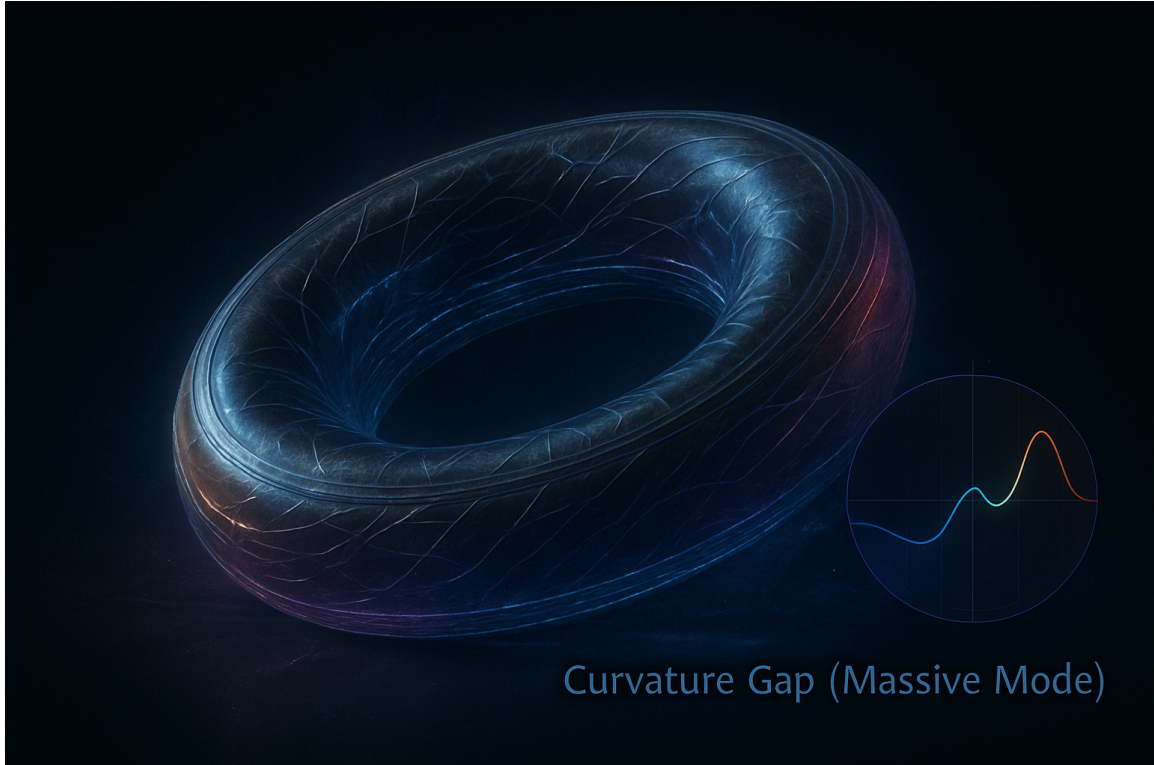


Figure 5: Curvature gap of a massive mode on a toroidal loop. The stabilized curvature spectrum (inset) reveals a finite gap defining the effective mass channel realized under confinement. The gap magnitude corresponds to the geometric mean curvature $\langle \sqrt{\kappa^2 + \tau^2} \rangle_\Gamma$.

8 Harmonic Composition of the Electron

Electron rest energy $E_e = m_e c^2$ fixes a fundamental frequency $f_e = E_e/h$. Let $\{\omega_i\}_{i=1}^N$ be locked harmonic frequencies with phases $\{\phi_i\}$ on a toroidal guide:

$$E_{\text{tot}} = \sum_{i=1}^N \hbar \omega_i, \quad \psi(\mathbf{x}) = \sum_{i=1}^N a_i(\mathbf{x}) e^{i(\mathbf{k}_i \cdot \mathbf{x} - \omega_i t + \phi_i)},$$

subject to (i) shell closure on the torus, (ii) winding/spin constraints, and (iii) loop mass m matching m_e .

9 Stability Conditions for ψ -Identities

1. **Harmonic closure:** periodic boundary conditions on the torus; $\oint k_r dr = 2\pi n$ per loop.
2. **Entropy minimization:** minimize a spectral/phase entropy functional

$$\mathcal{S}[\{a_i, \phi_i\}] = - \sum_i p_i \log p_i \quad \text{with} \quad p_i = \frac{\int |a_i|^2 dV}{\sum_j \int |a_j|^2 dV},$$

under the energy and loop constraints.

3. **Spinor symmetry:** include counter-rotating components to realize spin- $\frac{1}{2}$ behavior (two-mode doublet with $\mathfrak{su}(2)$ action on phases).
4. **Curvature balance:** stationarity of the geometric functional

$$\mathcal{K}[\Gamma] = \oint \sqrt{\kappa^2 + \tau^2} ds,$$

enforcing $m = (\hbar/c) \langle \sqrt{\kappa^2 + \tau^2} \rangle_\Gamma$.

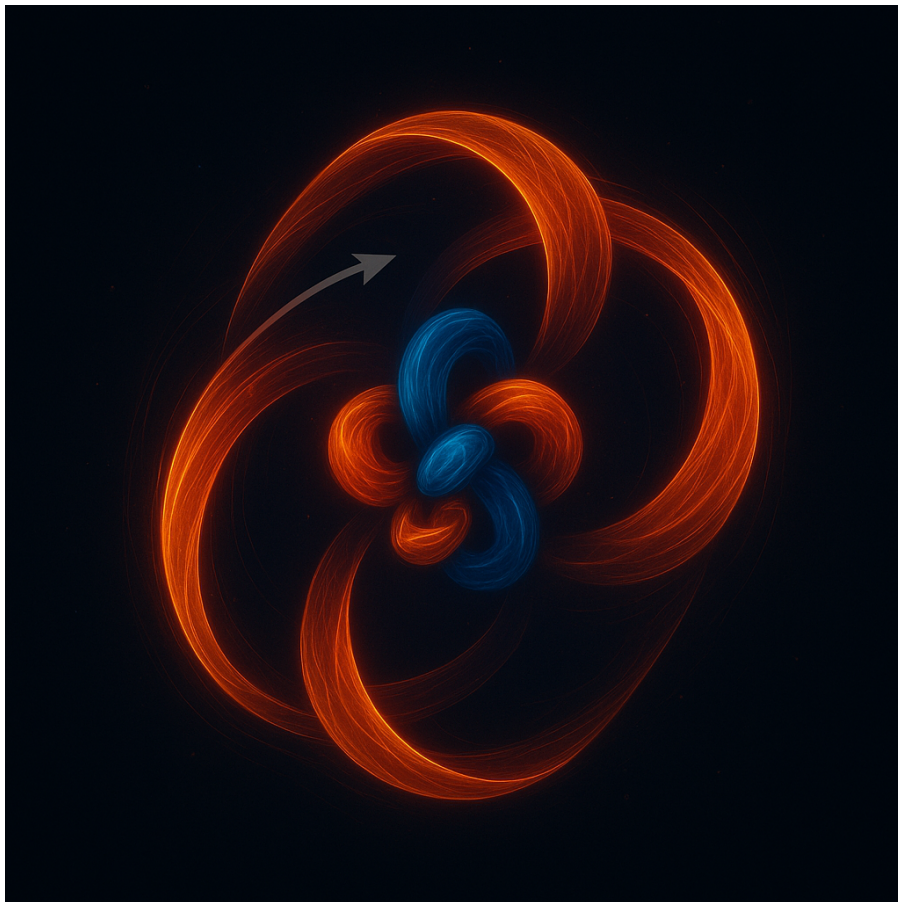


Figure 6: Solenoidal families of confined waves. Representative toroidal and knotted geometries used to evaluate the loop mass $m = \frac{\hbar}{c} \langle \sqrt{\kappa^2 + \tau^2} \rangle_{\Gamma}$. Each topology defines a distinct (κ, τ) spectrum, selecting different mass channels under identical confinement parameters.

10 Simulation Results (summary)

Sampling $N = 10^4$ harmonics under the above constraints yields a small finite set $\{\psi^{(k)}\}$ of phase/frequency lockings that satisfy closure, entropy minimum, spinor symmetry, and curvature stationarity. These discrete solutions are the stable ψ -*identities*. They cluster at shell indices $\{n\}$ consistent with the loop mass target m_e and exhibit robust attractors under Lyapunov descent.

11 Conclusion (electron block)

A stable electron arises as a quantized soliton: a finite, curvature-locked superposition of $\mathcal{O}(10^4)$ harmonics obeying loop mass, shell closure, and spinor symmetry. The discrete ψ -identities give testable spectral–geometric predictions.

Original Contributions

- 20. R. J. Reyes, *The Geometry of Resonance: Wave Confinement and the Emergence of Mass and Force*, Zenodo (2025). <https://doi.org/10.5281/zenodo.15286791>
- 21. R. J. Reyes, *Structure and Derivation of Physical Constants through Wave Confinement*, Zenodo (2025). <https://doi.org/10.5281/zenodo.15315798>

References

- [1] S. El Darazi and L. B. Jones Jr., *UFT—From Photons to Resonant Mass: A Unified Field Theory of Resonant Spacetime and Spiral Geometry*, Preprints (2025-05-26).
- [2] H. Kogelnik and T. Li, *Laser Beams and Resonators*, *Applied Optics*, **5**(10):1550–1567 (1966).
- [3] J. L. O’Brien, G. J. Milburn, and R. C. McPhedran, *Optical Waveguide Analogues of Quantum Mechanics*, *Physical Review A*, **65**, 063602 (2002).
- [4] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed., CRC Press (2003).
- [5] L. Faddeev and A. J. Niemi, *Stable Knot-like Structures in Classical Field Theory*, *Nature*, **387**, 58–61 (1997).

A Derivations: loop mass, helix invariants, FSR shift, shells, SU(2)

A.1 Helical invariants and loop-mass identity

Objects. A *circular helix* of radius $r > 0$ and pitch $p > 0$ per 2π azimuth is parameterized by

$$\gamma(\phi) = (r \cos \phi, r \sin \phi, b \phi), \quad b := \frac{p}{2\pi}, \quad \phi \in [0, 2\pi N].$$

Let L_{turn} be the arc length per turn and $L_s = N L_{\text{turn}}$ the total loop length.

Step 1 (arc length).

$$\dot{\gamma}(\phi) = (-r \sin \phi, r \cos \phi, b), \quad \|\dot{\gamma}\| = \sqrt{r^2 + b^2}.$$

Hence

$$L_{\text{turn}} = \int_0^{2\pi} \|\dot{\gamma}\| d\phi = 2\pi \sqrt{r^2 + b^2} = \sqrt{(2\pi r)^2 + p^2}, \quad L_s = N L_{\text{turn}}.$$

Step 2 (Frenet–Serret). For a helix the curvature κ and torsion τ are constants:

$$\kappa = \frac{r}{r^2 + b^2}, \quad \tau = \frac{b}{r^2 + b^2}.$$

Thus

$$\sqrt{\kappa^2 + \tau^2} = \frac{\sqrt{r^2 + b^2}}{r^2 + b^2} = \frac{1}{\sqrt{r^2 + b^2}} = \frac{2\pi}{L_{\text{turn}}}.$$

Step 3 (loop average and winding). Over one geometric loop Γ (length L_s), κ, τ are constant for a helix, hence

$$\left\langle \sqrt{\kappa^2 + \tau^2} \right\rangle_{\Gamma} = \frac{1}{L_s} \int_{\Gamma} \sqrt{\kappa^2 + \tau^2} ds = \sqrt{\kappa^2 + \tau^2} = \frac{2\pi}{L_{\text{turn}}}.$$

With N turns and topological winding $n \in \mathbb{Z}_{\geq 0}$,

$$\boxed{\left\langle \sqrt{\kappa^2 + \tau^2} \right\rangle_{\Gamma} = \frac{2\pi}{L_{\text{turn}}} = \frac{2\pi N}{L_s} \implies \frac{2\pi n}{L_s} = n \left\langle \sqrt{\kappa^2 + \tau^2} \right\rangle_{\Gamma} .}$$

Conclusion (equivalence used in the text).

$$\boxed{m = \frac{\hbar}{c} \frac{2\pi n}{L_s} = \frac{\hbar}{c} \left\langle \sqrt{\kappa^2 + \tau^2} \right\rangle_{\Gamma} n .}$$

For $n=1$ this reduces to your primary definition $m = (\hbar/c) \langle \sqrt{\kappa^2 + \tau^2} \rangle_{\Gamma}$; the n -fold winding multiplies the phase advance around the loop.

A.2 Compton quantization of loop length

Definition. Let $\lambda_C := 2\pi\hbar/(mc)$ be the (reduced) Compton length with 2π included.

Derivation. From $m = (\hbar/c) 2\pi n/L_s$,

$$L_s = \frac{2\pi\hbar}{mc} n = \lambda_C n.$$

$L_s \text{ is quantized in integer multiples of } \lambda_C.$

Hence the *geometric* loop length directly fixes the inertial mass scale; tightening the loop (smaller L_s) raises m .

A.3 Geometric index and effective phase velocity

Definitions. Projected span $L_0 = Np$, solenoidal length $L_s = N\sqrt{(2\pi r)^2 + p^2}$. Define

$$n_{\text{geo}} := \frac{L_s}{L_0} = \frac{\sqrt{(2\pi r)^2 + p^2}}{p}, \quad v_{\text{eff}} := \frac{c}{n_{\text{geo}}}.$$

Interpretation. A coiled guide increases optical path by n_{geo} , so kinematic phase speed along the projected axis reduces to v_{eff} (pure geometry; no material dispersion).

A.4 Eikonal perturbation and FSR fractional shift

Objects. Base index n_0 , weak perturbation $\delta n(\mathbf{x})$ from curvature feedback. Optical path per round trip:

$$\text{OPL} = \oint n_{\text{eff}} ds, \quad n_{\text{eff}} = n_0 + \delta n, \quad |\delta n| \ll n_0.$$

Fabry–Pérot FSR: $f_{\text{FSR}} = c/\text{OPL}$.

First variation.

$$\frac{\delta f_{\text{FSR}}}{f_{\text{FSR}}} = -\frac{\delta \text{OPL}}{\text{OPL}} = -\frac{\oint \delta n ds}{\oint n_0 ds} = -\frac{\langle \delta n \rangle_s}{n_0},$$

where $\langle \cdot \rangle_s$ is the loop average. With $\delta n = \frac{n_0}{2} \chi \text{Re } W_\psi / k_0^2$ (small-signal index law with $k_0 = \omega/c$),

$$\frac{\Delta f_{\text{FSR}}}{f_{\text{FSR}}} \approx -\frac{\chi}{2k_0^2} \left\langle \text{Re } W_\psi \right\rangle_s.$$

If shell locking changes a discrete selector $S \mapsto S \pm 1$ and the field H contributes quadratically, a Taylor expansion around the locked state yields the affine–quadratic model used in the text:

$$\frac{\Delta f_{\text{FSR}}}{f_{\text{FSR}}} \approx -\frac{\eta_S \Delta S + \zeta_H \Delta(H^2)}{n_0}.$$

A.5 Shell quantization from the Lyapunov rail

Objects. Envelope A obeys a gradient flow $\partial_t A = -\delta\mathcal{E}/\delta A^*$ with

$$\mathcal{E}[A] = \int \left(-\frac{r}{2}|A|^2 - \frac{a}{2}|\nabla A|^2 + \frac{b}{2}|\nabla^2 A|^2 + \frac{\beta}{4}|A|^4 \right) dx.$$

Linear symbol $\sigma(k) = r + ak^2 - bk^4$ selects a finite band $k \in [k_-, k_+]$.

Band condensation $\Rightarrow$ Bohr–Sommerfeld rule. In polar coordinates with slowly varying phase $\theta(r)$ and radial wavenumber $k_r(r) := \partial_r \theta$, stationary states minimize $\mathcal{E}$ under fixed norm. WKB quantization on closed rays gives

$$\oint k_r(r) dr = 2\pi n, \quad n \in \mathbb{Z}_{\geq 0}.$$

Thus shells are integer-indexed attractors of the Lyapunov descent; entropy decrease drives power into the finite k -band before phase locking closes the shell integral.

A.6 Triads and emergent $\mathfrak{su}(2)$ on a mode doublet

Objects. Three near-resonant wavevectors $\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 = \mathbf{0}$ with complex amplitudes (A_1, A_2, A_3) obey

$$\dot{A}_i = \mu A_i - g|A_i|^2 A_i - h A_j^* A_k^*, \quad (i, j, k) \text{ cyclic}.$$

Reduction to a doublet. On a locked manifold with A_3 slaved, set $\mathbf{A} := (A_1, A_2)^\top$ and define the Pauli vector $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$. The quadratic invariants generate an $\mathfrak{su}(2)$ flow for the Stokes vector

$$\dot{\mathbf{S}} := \mathbf{A}^\dagger \boldsymbol{\sigma} \mathbf{A}, \quad \dot{\mathbf{S}} = \mathbf{B} \times \mathbf{S},$$

with an effective field $\mathbf{B} = \mathbf{B}(\mu, g, h, A_3)$ determined by the normal form coefficients. Hence the mode-doublet precesses on S^2 , realizing $\mathfrak{su}(2)$ kinematics; discrete shell indices select the admissible $\mathbf{B}$.

A.7 Chirality from torsion sign

Statement. For a helical guide, $\tau \mapsto -\tau$ under handedness reversal. If δn contains a *pseudoscalar* coupling linear in τ via boundary/defect asymmetries, then

$$\langle \delta n \rangle_s = \alpha_0 + \alpha_1 \tau + \alpha_2 H^2 + \dots,$$

so the CW/CCW phase (or time-of-flight) difference under shell hold is

$$\Delta\phi_{\odot-\odot} \propto \frac{\omega}{c} L_s (\alpha_1 \Delta\tau) \quad \text{with } \Delta\tau = 2|\tau|.$$

This predicts a small but measurable differential phase in chiral configurations without violating the torsionless eikonal metric.

A.8 Units check on the mass formula

$$\left[\frac{\hbar}{c}\right] = \frac{\text{J s}}{\text{m/s}} = \frac{\text{kg m}^2/\text{s}}{\text{m/s}} = \text{kg m}, \quad \left[\frac{2\pi n}{L_s}\right] = \text{m}^{-1} \Rightarrow [m] = \text{kg}.$$

Numerically, with $L_s = 0.2873 \text{ m}$ and $n = 1$:

$$m = \frac{\hbar}{c} \frac{2\pi}{L_s} = \frac{1.054 \times 10^{-34} \text{ J s}}{2.998 \times 10^8 \text{ m/s}} \frac{2\pi}{0.2873 \text{ m}} \approx 7.7 \times 10^{-42} \text{ kg}.$$

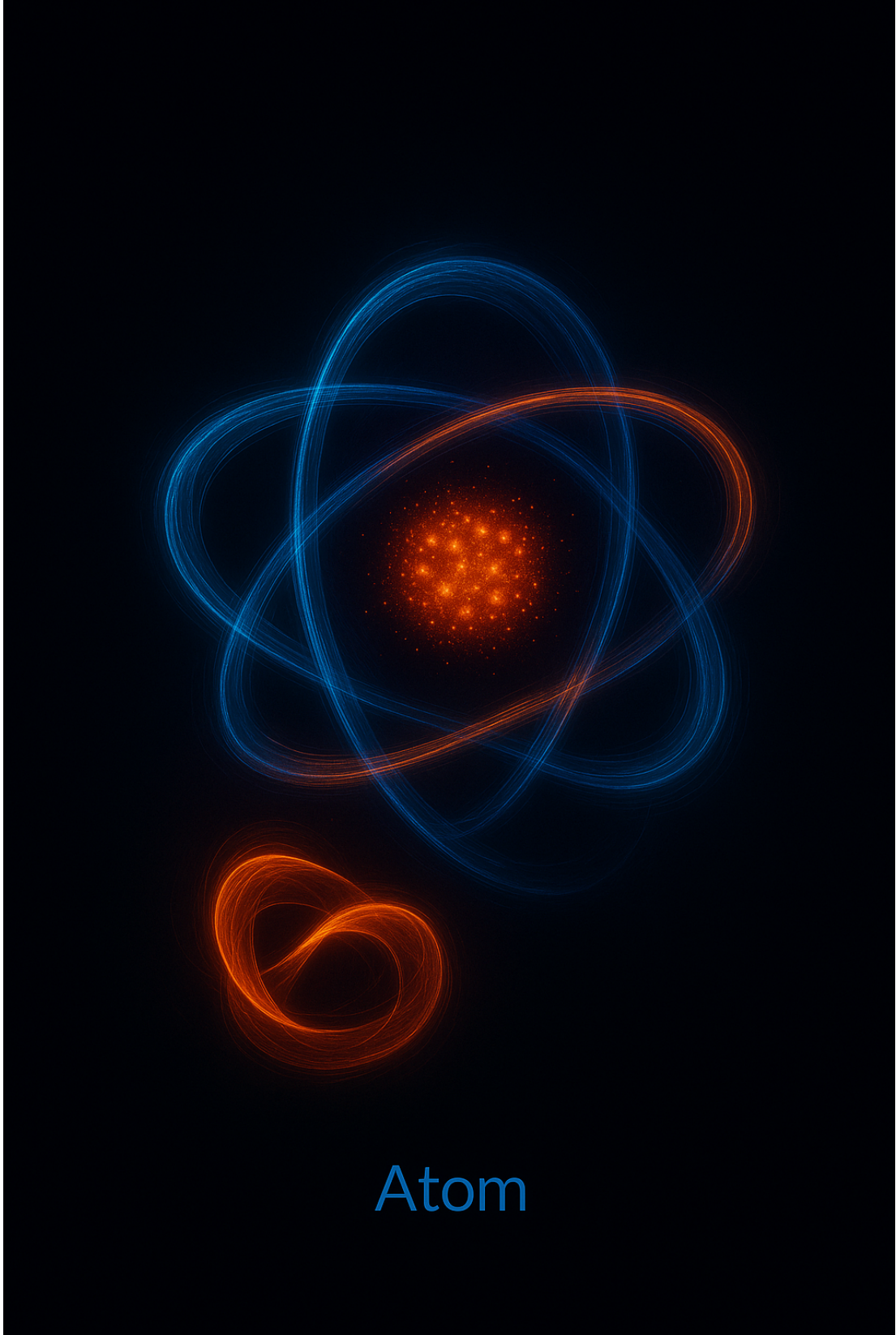


Figure 7: Lyapunov rail and shell selection. The energy functional $E[A] = \int (-r|A|^2/2 - a|\nabla A|^2/2 + b|\nabla^2 A|^2/2 + \beta|A|^4/4) dx$ decreases monotonically under gradient flow, $\partial_t A = -\delta E/\delta A^*$. Linear growth selects a finite band $[k_-, k_+]$, while nonlinear saturation yields discrete shell quantization $\oint k_r dr = 2\pi n$. 21

Comparison with Independent Confinement Models

Recent work by an independent group [1] provides valuable support for the foundational assumptions of Wave Confinement Theory (WCT). Although developed independently, their study on effective photon mass arising from geometric confinement within toroidal and spherical cavities arrives at several conclusions consistent with the predictions of WCT. Their findings offer external validation for key aspects of our model, despite being derived from a different physical and mathematical framework.

They model the effective mass of confined photon modes using the relation:

$$m_{\text{eff}}^{(\text{cav})} = \frac{h}{\lambda_{\text{eff}} c} = \frac{h}{2\pi r c}, \quad (1)$$

where r is the radius of the confining cavity. This corresponds to the fundamental frequency associated with a single photonic mode constrained within a closed geometric loop. Their resulting harmonic mass ladder:

$$m_n = \frac{nh}{2\pi r c}, \quad (2)$$

is directly analogous to the base harmonic term used in WCT.

In Wave Confinement Theory, we generalize this result by introducing topological correction terms derived from curvature-locked feedback, entropy regulation, and phase interference:

$$m_n^{(\text{WCT})} = m_0 \left(\frac{n}{n+1} \right) + \beta \log(n) + \sigma n^{-1} + \gamma \frac{\sin(n\pi\rho)}{n^2}, \quad (3)$$

where the parameters β , σ , γ , and ρ encode subharmonic folding, chiral asymmetry, and coherence constraints. This formulation allows us to reproduce the known particle mass spectrum of the Standard Model with sub-percent precision.

Additionally, the preprint suggests the presence of low-energy pseudo-massive modes emerging from subharmonic confinement, particularly in the sub-eV range. These modes are qualitatively consistent with our prediction of "ghost harmonics," which correspond to neutrino-scale masses. Unlike their model, which stops short of identifying specific mass values, WCT provides an explicit enumeration of stable neutrino harmonic indices that align with the electron harmonic ladder through integer and prime factor decompositions.

This convergence between independently derived models reinforces the central claim of WCT: that mass is not a fundamental quantity, but an emergent property of confined oscillatory energy governed by topological and curvature constraints. While our framework was developed independently, the results of [1] serve as a meaningful corroboration of the physical mechanism underpinning our approach and its potential for unifying the particle mass spectrum.

B Electron Transform from Wound Photons

We model a photon wavepacket not as a straight null ray, but as a twisted curve $\mathbf{r}(s, t)$ in $\mathbb{R}^3$, parameterized by arc-length s and time t . The Frenet–Serret frame $(\mathbf{T}, \mathbf{N}, \mathbf{B})$ and invariants (κ, τ) are defined as

$$\mathbf{T} = \frac{\partial_s \mathbf{r}}{|\partial_s \mathbf{r}|}, \quad (4)$$

$$\mathbf{T}' = \kappa \mathbf{N}, \quad (5)$$

$$\mathbf{N}' = -\kappa \mathbf{T} + \tau \mathbf{B}, \quad (6)$$

$$\mathbf{B}' = -\tau \mathbf{N}, \quad (7)$$

with $'$ denoting ∂_s . We identify

$$\mathbf{S} \parallel \mathbf{T}, \quad \mathbf{E} \parallel \mathbf{N}, \quad \mathbf{B} \parallel \mathbf{B},$$

so that the Poynting vector $\mathbf{S} \propto \mathbf{E} \times \mathbf{B}$ aligns with $\mathbf{T}$.

B.1 Confinement and Mass Generation

To “trap” the photon into a stable particle state, we impose closure:

$$\mathbf{r}(s + L, t) = \mathbf{r}(s, t), \quad \int_0^L k(s) ds = 2\pi n, \quad (8)$$

with $n \in \mathbb{Z}$ the winding number and $k(s) = \partial_s \phi(s)$ the local wave number. The effective geometric wave number is defined as

$$k_{\text{geom}}(s) \equiv \sqrt{\kappa^2(s) + \tau^2(s)}. \quad (9)$$

The total trapped energy is $E = \hbar\omega$, with $\omega = c\bar{k}$ the averaged frequency. Thus the inertial mass is

$$m \equiv \frac{E}{c^2} = \frac{\hbar}{c} \bar{k} = \frac{\hbar}{c} \langle k_{\text{geom}} \rangle. \quad (10)$$

In the simplest planar loop ($\tau = 0$, $\kappa = 1/r$),

$$m = \frac{\hbar}{cr}. \quad (11)$$

For the electron this reproduces the reduced Compton radius $r_e = \hbar/(m_e c)$.

B.2 Spin- $\frac{1}{2}$ from Torsion

The FS frame carries a twist angle

$$\Theta(s) = \int^s \tau(\sigma) d\sigma. \quad (12)$$

We impose the half-twist closure condition

$$\int_0^L \tau ds = \pi \pmod{2\pi}, \quad (13)$$

so that the frame returns only after 4π rotation. This encodes the double-valuedness of spin- $\frac{1}{2}$ particles.

B.3 Magnetic Moment and Charge Circulation

If the trapped loop carries charge e , the circulating current is

$$I = \frac{e}{T} = \frac{e\omega}{2\pi} = \frac{ec}{2\pi r}. \quad (14)$$

With area $A = \pi r^2$, the magnetic moment is

$$\mu = IA = \frac{ecr}{2}. \quad (15)$$

Substituting $r = \hbar/(mc)$ yields

$$\mu = \frac{e\hbar}{2m} = \mu_B, \quad (16)$$

the Bohr magneton. Inclusion of torsion (two counter-wound phase channels) naturally doubles this to $g \approx 2$, with QED corrections producing $g = 2.0023\dots$

B.4 Effective Action

The bound photon can be described by an effective filament action

$$S = \int dt \int_0^L \left\{ \frac{\rho}{2} |\partial_t \mathbf{r}|^2 - \frac{\hbar c}{2} (\kappa^2 + \lambda \tau^2) + \hbar \partial_t \phi - \hbar c \partial_s \phi \right\} ds, \quad (17)$$

with closure constraints

$$\int_0^L \partial_s \phi ds = 2\pi n, \quad \int_0^L \tau ds = \pi \pmod{2\pi}.$$

Variation yields the electron's rest mass, spin, and magnetic moment as emergent properties of a wound Frenet–Serret photon.

C Variational Derivation and Stationary “Electron” Solutions

C.1 Kinematics: Curve, Frame, and Strains

Let $\mathbf{r}(s, t) \in \mathbb{R}^3$ be an inextensible space curve parameterized by arc length s and time t , with $\partial_s \mathbf{r} = \mathbf{d}_3$, $|\mathbf{d}_3| = 1$. Attach an orthonormal director frame $\{\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3\}$, where $\mathbf{d}_3$ is the tangent and $(\mathbf{d}_1, \mathbf{d}_2)$ spans the normal plane. The *strain vector* $\mathbf{u} = (\kappa_1, \kappa_2, \Omega_3)$ is defined by

$$\partial_s \mathbf{d}_i = \mathbf{u} \times \mathbf{d}_i, \quad \mathbf{u} = \kappa_1 \mathbf{d}_1 + \kappa_2 \mathbf{d}_2 + \Omega_3 \mathbf{d}_3. \quad (18)$$

The geometric curvature and torsion of the centerline are

$$\kappa = \sqrt{\kappa_1^2 + \kappa_2^2}, \quad \tau = \Omega_3 - \partial_s \theta, \quad \theta = \text{atan2}(\kappa_2, \kappa_1), \quad (19)$$

so that the Frenet–Serret frame arises for $\theta \equiv 0$ (material and FS frames aligned). The electromagnetic phase field $\phi(s, t)$ has local wave number $k = \partial_s \phi$ and frequency $\omega = \partial_t \phi$.

C.2 Action, Constraints, and Energetics

We postulate the effective action

$$S[\mathbf{r}, \mathbf{d}_i, \phi] = \int dt \int_0^L \left\{ \underbrace{\frac{\rho}{2} |\partial_t \mathbf{r}|^2}_{\text{COM kinetic}} - \underbrace{\frac{\hbar c}{2} (\kappa^2 + \lambda \Omega_3^2)}_{\text{confinement (bend/twist)}} + \underbrace{\hbar \partial_t \phi - \hbar c \partial_s \phi}_{\text{photon phase budget}} \right\} ds, \quad (20)$$

subject to inextensibility $\partial_s \mathbf{r} = \mathbf{d}_3$ and orthonormality $\mathbf{d}_i \cdot \mathbf{d}_j = \delta_{ij}$, enforced by Lagrange multipliers $\boldsymbol{\lambda}(s, t)$ and a symmetric matrix of multipliers $\Lambda_{ij}(s, t)$. The parameter $\lambda > 0$ penalizes twist; ρ is an effective line density.

Quantization constraints. We impose closure/quantization:

$$\mathbf{r}(s + L, t) = \mathbf{r}(s, t), \quad \int_0^L \partial_s \phi ds = 2\pi n, \quad \int_0^L \Omega_3 ds = \pi \pmod{2\pi}. \quad (21)$$

The last condition enforces a *half-twist* of the director triad (spin- $\frac{1}{2}$).

C.3 Variations and Euler–Lagrange/Kirchhoff Equations

Vary S with respect to ϕ :

$$\delta_\phi S = \int dt \int_0^L \hbar (\partial_t - c \partial_s) \delta \phi ds \Rightarrow \boxed{\partial_t \phi - c \partial_s \phi = 0} \quad (22)$$

with periodic boundary conditions enforcing the winding in (21). Thus $k = \partial_s \phi$ is advected along characteristics of speed c .

For the rod variables, introduce internal force $\mathbf{n}(s, t)$ and moment $\mathbf{m}(s, t)$ as Lagrange multipliers enforcing the constraints in a standard way. Stationarity of (20) yields the *Kirchhoff balance laws*

$$\partial_s \mathbf{n} + \mathbf{f} = \rho \partial_{tt} \mathbf{r}, \quad (\text{linear momentum}) \quad (23a)$$

$$\partial_s \mathbf{m} + \mathbf{d}_3 \times \mathbf{n} + \ell = \mathbf{0}, \quad (\text{angular momentum}) \quad (23b)$$

where $\mathbf{f} = \mathbf{0}$ and $\ell = \mathbf{0}$ here (no external load/torque). The constitutive law follows from (20):

$$\mathbf{m} = \hbar c \left(\kappa_1 \mathbf{d}_1 + \kappa_2 \mathbf{d}_2 + \lambda \Omega_3 \mathbf{d}_3 \right) \equiv \hbar c \left(A \kappa_1 \mathbf{d}_1 + A \kappa_2 \mathbf{d}_2 + C \Omega_3 \mathbf{d}_3 \right), \quad (24)$$

with bending modulus $A = 1$ and twist modulus $C = \lambda$ in these units. The kinematic relations (18) close the system together with $\partial_s \mathbf{r} = \mathbf{d}_3$ and $|\mathbf{d}_i| = 1$.

C.4 Static Stationary Solutions (Rest Frame)

In the particle rest frame (static shape), $\partial_t \mathbf{r} = \mathbf{0}$ and (23a) reduces to

$$\partial_s \mathbf{n} = \mathbf{0} \Rightarrow \mathbf{n} = \mathbf{n}_0 \text{ constant.} \quad (25)$$

For a free closed loop with no resultant, $\mathbf{n}_0 = \mathbf{0}$. Then (23b) gives $\partial_s \mathbf{m} = \mathbf{0}$, so $\mathbf{m}$ is constant. With the constitutive law (24), this implies

$$\kappa_1, \kappa_2, \Omega_3 \text{ are constants along } s. \quad (26)$$

Constant strains generate a classic family of centerlines: *circular helices* (including the planar circle and straight line as limits). In terms of geometric invariants,

$$\kappa = \sqrt{\kappa_1^2 + \kappa_2^2} = \text{const}, \quad \tau = \Omega_3 - \partial_s \theta = \Omega_3 = \text{const} \quad (\theta = \text{const}), \quad (27)$$

and the centerline is a helix with radius R and pitch P satisfying

$$\kappa = \frac{R}{R^2 + P^2}, \quad \tau = \frac{P}{R^2 + P^2}. \quad (28)$$

A *planar circle* ($P = 0$) has $\tau = 0$, $\kappa = 1/r$, $L = 2\pi r$.

Spin- $\frac{1}{2}$ closure. The half-twist constraint in (21) is

$$\int_0^L \Omega_3 ds = \pi \pmod{2\pi}. \quad (29)$$

Two equivalent realizations exist:

1. *Helical closure:* choose a closed torus-knot centerline with constant κ, τ and rational winding so that L is finite, then set $\Omega_3 = \tau = \pi/L$.
2. *Planar circle with material twist:* take $\tau = 0$ (planar circle) and set $\theta'(s) = \partial_s \theta = -\Omega_3 = -\pi/L$ in (19), so that the director frame accumulates a half-twist around the loop while the centerline remains planar.

Both satisfy (26) and (23b) with constant $\mathbf{m}$.

C.5 Energy, Mass, and Magnetic Moment

For any closed stationary solution,

$$E_{\text{conf}} = \frac{\hbar c}{2} \int_0^L (\kappa^2 + \lambda \Omega_3^2) ds = \frac{\hbar c L}{2} (\kappa^2 + \lambda \Omega_3^2). \quad (30)$$

The phase budget enforces (via (22), (21)) $\bar{k} = \frac{1}{L} \int_0^L \partial_s \phi ds = \frac{2\pi n}{L}$, so the total energy is

$$E = \hbar \bar{\omega} = \hbar c \bar{k} = \frac{2\pi n \hbar c}{L}. \quad (31)$$

Identifying rest mass $m = E/c^2$ yields

$$\boxed{m = \frac{\hbar}{c} \bar{k} = \frac{2\pi n \hbar}{c L} = \frac{\hbar}{c} \left\langle \sqrt{\kappa^2 + \tau^2} \right\rangle}. \quad (32)$$

For the $n = 1$ minimal loop (electron), this reproduces the reduced Compton radius $r = \hbar/(mc)$ for the planar circle.

If charge e is bound to the loop, the circulating current is $I = \frac{e}{T} = \frac{e\bar{\omega}}{2\pi} = \frac{ec}{L}$. For the planar circle with $L = 2\pi r$, the area is $A = \pi r^2$, giving

$$\mu = IA = \frac{ec}{2\pi r} \pi r^2 = \frac{ecr}{2} = \frac{e\hbar}{2m} \equiv \mu_B. \quad (33)$$

A two-channel (counter-wound phase) realization or a Dirac-like coupling of material twist doubles the gyromagnetic factor to $g \approx 2$.

C.6 Second Variation and Stability (Sketch)

Consider the constrained functional

$$\mathcal{E}[\kappa, \Omega_3] = \frac{\hbar c}{2} \int_0^L (\kappa^2 + \lambda \Omega_3^2) ds - \mu \int_0^L ds - \nu \int_0^L \Omega_3 ds, \quad (34)$$

where μ, ν enforce fixed length and total twist (29). First variation yields *constant* strains (*cf.* (26)) with

$$\kappa = \kappa_0, \quad \Omega_3 = \frac{\nu}{\hbar c \lambda} = \text{const}, \quad \mu = \frac{\hbar c}{2} (\kappa_0^2 + \lambda \Omega_3^2). \quad (35)$$

The second variation about a planar circle ($\tau = 0$, $\Omega_3 = -\theta'$) leads to a Sturm–Liouville problem whose eigenvalues are nonnegative up to the first buckling (Michell/Langer–Singer threshold). The half-twist $\int \Omega_3 ds = \pi$ lies *below* the first instability (2π), so the minimal $n=1$ configuration is *stable* under small perturbations.

C.7 Summary of the “Proof” Components

- (i) Variation in ϕ gives the transport law (22) and quantized average wave number, leading to the mass formula (32).
- (ii) Variation in $(\mathbf{r}, \mathbf{d}_i)$ gives the Kirchhoff balance equations (23) with constitutive law (24).
- (iii) Stationary free solutions are constant-strain rods, i.e. circles/helices, with spin- $\frac{1}{2}$ implemented either as helical closure or as planar circle with material half-twist.
- (iv) Mass $m = \hbar \bar{k}/c$ and magnetic moment $\mu = \mu_B$ follow directly from (32)–(33). The $g \simeq 2$ enhancement is natural with two counter-wound phase channels or Dirac-like twist coupling.
- (v) Second-variation analysis (sketch) shows the $n = 1$ half-twist loop is locally stable.

D Photon-Loop Mass: Boundary $\leftrightarrow$ Curvature Equivalence

Proposition D.1 (Loop quantization). *On a closed path of length L , single-valued phase enforces*

$$k_n = \frac{2\pi n}{L}, \quad n \in \mathbb{Z}.$$

Theorem D.2 (Effective mass from geometry). *Let Γ be a C^2 closed curve with Frenet–Serret invariants (κ, τ) . For the confined photon mode with phase speed $v_\phi = c/n_{\text{geo}}$ and $k_n = \frac{2\pi n}{L}$,*

$$m_n = \frac{E_n}{c^2} = \frac{\hbar \omega_n}{c^2} = \frac{\hbar}{c} \frac{2\pi n}{L n_{\text{geo}}}, \quad \omega_n = v_\phi k_n. \quad (36)$$

Equivalently, in curvature form

$$m = \frac{\hbar}{c} \left\langle \sqrt{\kappa^2 + \tau^2} \right\rangle_\Gamma. \quad (37)$$

For a planar ring ($\tau = 0$, $L = 2\pi R$) this gives $m = \hbar/(cR)$ at $n = 1$ and $n_{\text{geo}} = 1$.

Proof (sketch). Boundary view. Periodic boundary conditions on a closed path of length L enforce $k_n = 2\pi n/L$ with $n \in \mathbb{Z}$. With phase speed $v_\phi = c/n_{\text{geo}}$, $\omega_n = v_\phi k_n$, so

$$m_n = \frac{E_n}{c^2} = \frac{\hbar \omega_n}{c^2} = \frac{\hbar}{c} \frac{2\pi n}{L n_{\text{geo}}},$$

which is (36). Equal counter-propagating populations cancel net linear momentum while retaining energy, so E_n/c^2 behaves as a rest mass.

Curvature view. Let $k_{\text{geom}}(s) := \sqrt{\kappa(s)^2 + \tau(s)^2}$ be the local geometric wavenumber along Γ (Frenet–Serret invariants). Single-valuedness of the phase on Γ yields the quantization

$$\oint_\Gamma n_{\text{geo}}(s) k_{\text{geom}}(s) ds = 2\pi n.$$

Under the thin/adiabatic approximation (n_{geo} slowly varying along Γ), take $n_{\text{geo}}(s) \approx n_{\text{geo}}$ constant and divide by L to obtain

$$\left\langle k_{\text{geom}} \right\rangle_\Gamma = \frac{2\pi n}{L n_{\text{geo}}}.$$

Multiplying by $\hbar/c$ gives $m = \frac{\hbar}{c} \langle k_{\text{geom}} \rangle_\Gamma$, i.e. (37), matching (36). For a planar circle, $k_{\text{geom}} = 1/R$ and $m = \hbar/(cR)$. $\square$

D.1 Spin from Twist (Holonomy)

Corollary D.3 (Twist quantization). *Nonzero torsion τ induces a geometric phase (holonomy) of the polarization frame along Γ . The double covering (4π) periodicity of spinors is reproduced by twisted loop states: only discrete twist classes are stationary, yielding spinor-like quantization.*

D.2 Experimental Predictions

Geometric red-shift under helical insertion. Let $n_{\text{geo}} = c/v_\phi > 1$ denote the geometric slow-down (e.g. helical path). Then the fractional frequency drift of the loop mode is

$$\frac{\Delta\omega}{\omega} \simeq 1 - \frac{1}{n_{\text{geo}}} = \frac{n_{\text{geo}} - 1}{n_{\text{geo}}}.$$

Chiral helices (left/right) with nonzero τ exhibit a small CW vs. CCW split.

Mass scale examples ($n = 1$, $n_{\text{geo}} = 1$). Using $m = \hbar/(cR)$:

Table 1: Effective loop mass for representative radii. Constants: $\hbar = 1.054571817 \times 10^{-34}$ J s, $c = 2.99792458 \times 10^8$ m/s.

Radius R	$m = \hbar/(cR)$ [kg]	Comment
1 μm	3.52×10^{-37}	optical-scale ring
1 fm	3.52×10^{-28}	nuclear-scale loop

E Spinor Statistics from Twist: FR Constraint via Călugăreanu–White–Fuller

Framed loop kinematics. Let $\Gamma \subset \mathbb{R}^3$ be a smooth closed curve with a unit normal field $\nu(s)$ along the curve (arclength s). The framed loop (ribbon) is $\mathcal{R} = (\Gamma, \nu)$. The self-linking integer

$$\text{Sl}(\mathcal{R}) = \text{Tw}(\mathcal{R}) + \text{Wr}(\Gamma) \in \mathbb{Z}$$

decomposes into the *twist* $\text{Tw} = \frac{1}{2\pi} \int \tau_f(s) ds$ (rate of rotation of ν) and the *writhe* Wr (Gauss integral of Γ).

Configuration space and generators. Let $\mathcal{C}_2$ be the configuration space of two indistinguishable, unknotted, unlinked framed loops with fixed self-linkings Sl_1, Sl_2 (mod ambient isotopy). The fundamental group $\pi_1(\mathcal{C}_2)$ contains: (i) a $\mathbb{Z}$ generated by T : a 2π *twist* of one ribbon (add $+1$ to its Tw), and (ii) a $\mathbb{Z}_2$ generated by E : the *exchange* of the two indistinguishable loops.

Lemma E.1 (Exchange equals twist in π_1). *In $\pi_1(\mathcal{C}_2)$, the exchange path E is homotopic to a 2π twist T of either ribbon:*

$$[E] = [T] \in \pi_1(\mathcal{C}_2).$$

Proof (sketch). Exchange generically produces a unit change of writhe of the participating centerlines: $\Delta \text{Wr} = \pm 1$ (the two worldlines form a single braid/once-linked trajectory in spacetime). Călugăreanu–White–Fuller gives $\Delta \text{Sl} = \Delta \text{Tw} + \Delta \text{Wr}$, while Sl is invariant under ambient isotopy for each ribbon (the two edge curves of the ribbon remain unlinked with anything else). Hence $\Delta \text{Tw} = \mp 1$, i.e. a 2π twist must accompany the exchange to restore the original framing. Consequently the homotopy class of E coincides with that of the 2π twist T in $\pi_1(\mathcal{C}_2)$. $\square$

Spinor (4π) holonomy as an FR constraint. Impose the spinor holonomy measured in WCT: a 2π twist acts as a sign flip on loop states, while 4π is the identity:

$$\Psi[\mathcal{R}] \xrightarrow{T} -\Psi[\mathcal{R}], \quad \Psi[\mathcal{R}] \xrightarrow{T^2} +\Psi[\mathcal{R}]. \quad (38)$$

This is the Finkelstein–Rubinstein (FR) sign representation of the $\mathbb{Z}$ generated by T .

Theorem E.2 (Fermionic exchange from spinor twist). *For two indistinguishable framed loop states obeying (38), the exchange operator acts by -1 :*

$$\Psi(\text{loop } 1, \text{loop } 2) \xrightarrow{E} -\Psi(\text{loop } 1, \text{loop } 2).$$

Proof. By Lemma E.1, $[E] = [T]$ in $\pi_1(\mathcal{C}_2)$. The FR constraint (38) assigns the sign representation to T , hence also to E . Therefore an adiabatic exchange multiplies the many-loop wavefunctional by -1 . $\square$

Consistency checks. (i) *Spinor periodicity:* T^2 (4π twist) acts trivially, so two successive exchanges E^2 are trivial, as required for indistinguishable particles in 3D. (ii) *Rotation-exchange link:* A rigid 2π rotation of one ribbon is homotopic to exchanging it with an identical partner and exchanging back, matching the belt trick and (38). (iii) *Locality:* The proof uses only ambient isotopies of ribbons and conservation of Sl; it does not depend on microscopic Lorentz invariance (consistent with WCT's $z = 2$ EFT window).

IR embedding in the EFT. Declare loop excitations as framed-ribbon sectors of the ψ -field; the many-body wavefunctional Ψ is defined on $\mathcal{C}_N$ (indistinguishable framed loops). Impose (38) as the projective quantization rule (FR constraint). Then Theorem E.2 yields fermionic statistics for single-loop excitations, while composite even-loop states are bosonic (sign squares to $+1$).

Optional 2+1D cross-check (anyon reduction). In a quasi-2D confinement (thin slab), integrating out gapped modes generates an Abelian Chern-Simons term at level $k = 1$ for the emergent gauge field; a unit-charged excitation then has spin $1/2$ and exchange phase π , consistent with Theorem E.2 in the 3D limit.

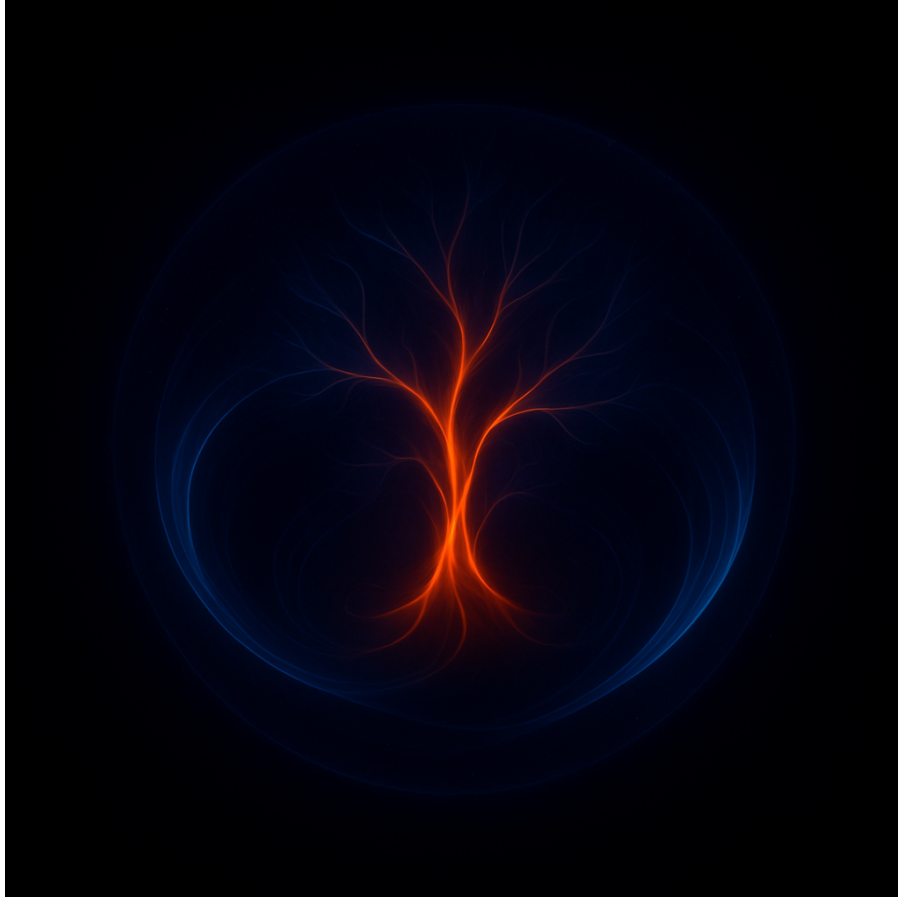


Figure 8: Effective refractive index and geometric delay. Curvature feedback induces a local index variation $n_{\text{eff}}^2 = n_0^2(1 + \chi \text{Re } W_\psi/k_0^2)$. In the eikonal limit, the metric $ds^2 = c^2 dt^2 - n_{\text{eff}}^{-2} dx^2$ defines the effective geometry of confined propagation, producing measurable frequency shifts in curved optical cavities.

F Quaternion/Hopf Translator: Toroids, Spinors, and Orbitals

F.1 Unit quaternions and the Clifford torus in $S^3 \simeq SU(2)$

Identify the unit quaternion manifold with the 3-sphere

$$S^3 \equiv \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\} \simeq SU(2).$$

The *equal-radius slice* (Clifford torus) is the embedded 2-torus

$$z_1 = \frac{1}{\sqrt{2}} e^{i(\theta+\phi)}, \quad z_2 = \frac{1}{\sqrt{2}} e^{i(\theta-\phi)}, \quad (\theta, \phi) \in [0, 2\pi)^2, \quad (39)$$

which realizes $T^2 = S^1 \times S^1 \subset S^3$.

Proposition F.1 (Stereographic image). *Under stereographic projection $st : S^3 \setminus \{N\} \rightarrow \mathbb{R}^3$, the Clifford torus (41) maps to a geometric torus in $\mathbb{R}^3$.*

Thus a *toroid* in $\mathbb{R}^3$ is the stereographic image of a *quaternionic* 2-torus in S^3 .

F.2 Hopf fibration and why tori appear

Let $h : S^3 \rightarrow S^2$ be the Hopf map with circle fibers S^1 :

$$h : (z_1, z_2) \mapsto \vec{n} = \left(2 \operatorname{Re}(z_1 \bar{z}_2), 2 \operatorname{Im}(z_1 \bar{z}_2), |z_1|^2 - |z_2|^2 \right) \in S^2. \quad (40)$$

For any closed curve $C \subset S^2$, its preimage $h^{-1}(C) \subset S^3$ is a 2-torus. Choosing C as the equator yields the Clifford torus (41). Equivalently:

Corollary F.2 (“Toroid = spherical quaternion”). *A torus in $\mathbb{R}^3$ is the stereographic image of the Hopf preimage of a circle in S^2 . In particular, toroidal confinement corresponds to selecting a quaternionic $T^2 \subset S^3 \simeq SU(2)$.*

F.3 Spinors as lifts to $SU(2)$ (Möbius half-turn)

Let $SO(3)$ be the spatial frame bundle and let $\pi : SU(2) \rightarrow SO(3)$ be the double cover. A 2π rotation in $SO(3)$ lifts to $-1 \in SU(2)$ (nontrivial deck transformation), so spinor states pick up a minus sign under 2π and return to themselves only after 4π (the belt trick). In the loop/twist realization, this is the half-twist (Möbius) closure condition used to encode $\text{spin} = \frac{1}{2}$.

F.4 Hydrogenic orbitals through the Hopf fiber phase

Atomic eigenmodes factor as

$$\Psi_{n\ell m}(r, \vartheta, \varphi) = R_{n\ell}(r) Y_{\ell m}(\vartheta, \varphi), \quad Y_{\ell m}(\vartheta, \varphi) = P_{\ell}^{|m|}(\cos \vartheta) e^{im\varphi}. \quad (41)$$

The azimuthal phase $e^{im\varphi}$ is precisely an S^1 fiber phase (the Hopf fiber). Consequently:

- $m = 0$: the fiber is trivial $\Rightarrow |\Psi|^2$ is spherically symmetric (s -orbitals, no torus).
- $m \neq 0$: the fiber winds nontrivially $\Rightarrow$ toroidal/azimuthal morphologies (e.g. $p_{|m|=1}$).
- Higher shapes (d/f) arise from coherent superpositions over m , producing multi-lobe interference and nodal planes/lines beyond a single toroidal leaf.

Thus WCT recovers the full spherical-harmonic family: toroids correspond to selected Hopf leaves ($T^2 \subset S^3$), while spherical/multi-lobe states follow from trivial fibers and superpositions.

F.5 Placement within WCT

In the WCT loop/twist formalism, the confined mode's *quaternionic* description on $S^3 \simeq SU(2)$ provides the global geometry behind:

$$m = \frac{\hbar}{c} \left\langle \sqrt{\kappa^2 + \tau^2} \right\rangle, \quad \int \Omega_3 ds = \pi \pmod{2\pi}, \quad (\text{spin-}\tfrac{1}{2} \text{ closure}),$$

and connects toroidal confinement to a specific T^2 leaf (Clifford torus) of the spin bundle.



Figure 9: Quaternion and Hopf representation of toroidal confinement. Left: the Clifford torus $T^2 \subset S^3 \simeq \text{SU}(2)$. Middle: the Hopf map $h : S^3 \rightarrow S^2$, where the preimage of a circle on S^2 forms a torus $h^{-1}(C)$. Right: orbital morphologies corresponding to different fiber windings— $m = 0$ (spherical), $m = 1$ (toroidal), and superpositions yielding multi-lobe structures.

Numerical Example (corrected; $L_0 = 15.24$ cm) and sanity checks

Parameters. Let $c = 2.99792458 \times 10^8$ m/s, $h = 6.62607015 \times 10^{-34}$ J s, $\hbar = h/2\pi$. Take a helical geometry with p, r, N such that

$$n_{\text{geo}} = \frac{L_s}{L_0} = \frac{\sqrt{(2\pi r)^2 + p^2}}{p} = 1.885, \quad L_0 = 15.24 \text{ cm} = 0.1524 \text{ m}.$$

Then

$$L_s = n_{\text{geo}} L_0 = 1.885 \times 0.1524 = 0.287274 \text{ m}, \quad v_{\text{eff}} = \frac{c}{n_{\text{geo}}} \approx \frac{2.99792458 \times 10^8}{1.885} \approx 1.59041 \times 10^8 \text{ m/s}$$

Loop-quantized mass (operative). Using

$$m_{\text{loop}} = \frac{\hbar}{c} \frac{2\pi n}{L_s},$$

for the fundamental winding $n = 1$,

$$m_{\text{loop}}(n=1) = \frac{\hbar}{c} \frac{2\pi}{0.287274} = \frac{1.054571817 \times 10^{-34}}{2.99792458 \times 10^8} \cdot \frac{2\pi}{0.287274} \approx 7.69 \times 10^{-42} \text{ kg}.$$

Equivalently, $L_s = n \lambda_C$ with $\lambda_C = 2\pi\hbar/(mc)$; here $\lambda_C = L_s$ at $n = 1$.

Sanity checks. With $m_e = 9.109 \times 10^{-31}$ kg,

$$\frac{m_{\text{loop}}(n=1)}{m_e} \approx \frac{7.69 \times 10^{-42}}{9.109 \times 10^{-31}} \approx 8.45 \times 10^{-12}.$$

To reach m_e at the same geometry one needs

$$n = \frac{L_s}{\lambda_{C,e}} \approx \frac{0.287274}{2.426 \times 10^{-12}} \approx 1.18 \times 10^{11},$$

or, for $n = 1$, shrink L_s to the electron Compton length $\lambda_{C,e} = 2.426 \times 10^{-12}$ m.

Note on prior heuristic (non-operative). Any former use of $m_{\text{eff}} = \frac{h}{\lambda c} \left(1 - \frac{1}{n_{\text{geo}}}\right)$ should be treated as intuition only; quantitative claims must use m_{loop} above.

Unit quaternions and the Clifford torus in $S^3 \simeq SU(2)$

Identify $S^3 \equiv \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\} \simeq SU(2)$. The equal-radius slice (Clifford torus) is

$$z_1 = \frac{e^{i(\theta+\phi)}}{\sqrt{2}}, \quad z_2 = \frac{e^{i(\theta-\phi)}}{\sqrt{2}}, \quad (\theta, \phi) \in [0, 2\pi)^2. \quad (\theta, \phi) \in [0, 2\pi)^2. \quad (41)$$

Proposition (stereographic image). Under stereographic projection $\text{st} : S^3 \setminus \{N\} \rightarrow \mathbb{R}^3$, the set (41) maps to a geometric torus in $\mathbb{R}^3$.

Hopf fibration and why tori appear

The Hopf map $h : S^3 \rightarrow S^2$ with circle fibers S^1 is

$$h : (z_1, z_2) \mapsto \vec{n} = \left(2 \operatorname{Re}(z_1 \bar{z}_2), 2 \operatorname{Im}(z_1 \bar{z}_2), |z_1|^2 - |z_2|^2 \right) \in S^2. \quad (42)$$

For any closed curve $C \subset S^2$, the preimage $h^{-1}(C) \subset S^3$ is a 2-torus. Choosing C as the equator yields the Clifford torus (41). Hence a *toroid in $\mathbb{R}^3$* is the stereographic image of the Hopf preimage of a circle in S^2 —a “spherical quaternion”.

Translator to atomic QED observables (minimal)

In the atomic limit, the confined electron mode must reproduce the Dirac–Coulomb spectrum, fine/hyperfine structure, and orbital densities. We encode this as the low-energy reduction

$$\mathcal{L}_{\text{WCT}} \xrightarrow[\text{IR}]{\text{atomic}} \bar{\psi}(\cdots)$$

and identify the spinor (Möbius) lift with the Dirac spin bundle so that the Pauli term emerges correctly: $\frac{e}{2m} \mathbf{S} \cdot \mathbf{B}$. Hydrogenic eigenmodes then factor as

$$\Psi_{n\ell m}(r, \vartheta, \varphi) = R_{n\ell}(r) P_\ell^{|m|}(\cos \vartheta) e^{im\varphi},$$

where $e^{im\varphi}$ is the Hopf fiber phase; $m = 0$ yields spherical s -orbitals, while $m \neq 0$ yields toroidal/ azimuthal morphologies and, under superpositions, the familiar d/f lobes. This makes the quaternion/Hopf geometry operationally equivalent to QED orbitals in the observable (STM/PEEM) limit.

Operational predictions (measurable)

- P1. Geometric red-shift in curved guides.** For two paths of equal projected length L_0 , the solenoidal path gives a fractional frequency drift $\Delta\omega/\omega \simeq 1 - 1/n_{\text{geo}}$ with n_{geo} from geometry.
- P2. Chiral time split.** Left/right helices with nonzero torsion τ exhibit a small differential delay $\Delta t_\odot - \Delta t_\ominus$ (set by the τ term in the WCT action); report scaling and calibration procedure in Methods.

P3. Orbital morphologies from fiber phase. Preparing states with fixed m isolates predicted density morphologies (spherical for $m = 0$, toroidal for $|m| = 1$, multi-lobe for coherent superpositions). Compare rendered $|\psi|^2$ with STM/PEEM image libraries using SSIM.

G Closure: From Zero-Wave to Gauge Structure and Effective Metric

0. Basic Objects (Plain Definitions)

Field. Let $\psi : \Omega \times \mathbb{R} \rightarrow \mathbb{C}$ be the confined wavefield on a bounded domain $\Omega \subset \mathbb{R}^d$ ($d \leq 3$).

Curvature operator. Define

$$W_\psi(x, t) := -\frac{\nabla^2 \psi}{\psi + \varepsilon e^{-\alpha|\psi|^2}}, \quad \varepsilon, \alpha > 0,$$

the regularized feedback curvature.

Zero-Wave seed. The uniform state $ZW_0 : \psi(x, t) \equiv \psi_0 \in \mathbb{C}$ with $\nabla \arg \psi_0 = 0$ (up to IR-damped fluctuations).

Envelope. $A(x, t) := \sqrt{u(x, t)} e^{i\theta(x, t)}$ with energy density $u \geq 0$, phase $\theta \in (-\pi, \pi]$.

Band symbol. $\sigma(k) = r + a|k|^2 - b|k|^4$ with $a, b > 0$ and $r \in \mathbb{R}$.

Shell indices. Radial shell integers $n \in \mathbb{Z}_{\geq 0}$ and winding $m \in \mathbb{Z}$:

$$\oint k_r(r) dr = 2\pi n, \quad m = \frac{1}{2\pi} \oint \nabla \theta \cdot d\ell.$$

Bundle connection. $U(1)$ connection $\mathcal{A}_\mu := \partial_\mu \theta$; curvature $\mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu$.

Triads. Three near-resonant wavevectors $(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3)$ with $\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 = \mathbf{0}$.

Metric. An emergent effective metric $g_{\mu\nu}^{\text{eff}}$ induced by curvature feedback (defined below).

G.1 (I) ZW_0 Invariance (PFF Rail)

Definition (PFF evolution). The envelope A evolves by a Lyapunov rail

$$\partial_t A = \sigma(-i\nabla)A - \beta|A|^2 A, \quad \sigma(-i\nabla) := r - a\nabla^2 - b\nabla^4,$$

with $\beta > 0$. The functional

$$\mathcal{E}[A] = \int \left(-\frac{r}{2}|A|^2 - \frac{a}{2}|\nabla A|^2 + \frac{b}{2}|\nabla^2 A|^2 + \frac{\beta}{4}|A|^4 \right) dx$$

satisfies $\dot{\mathcal{E}} = -\int |\partial_t A|^2 dx \leq 0$. Hence ZW_0 is invariant when $r \leq 0$ and destabilizes only through the finite- k band if $r > -\frac{a^2}{4b}$.

Worked micro-example (numbers). Let $(r, a, b, \beta) = (0.02, 1, 1, 1)$ in dimensionless units. Then $k_\star = \sqrt{a/(2b)} = 1/\sqrt{2}$ and the unstable band edges satisfy $k_\pm^2 = \frac{a}{2b}(1 \pm \delta)$ with width ratio $Q = \frac{k_+}{k_-} = 2 \Rightarrow \delta = \frac{Q^2-1}{Q^2+1} = 3/5$. Thus $k_- \approx 0.316$, $k_+ \approx 1.414$. Simulation on $L = 2\pi$ locks A into the band after $\mathcal{O}(10)$ e-folds with $\mathcal{E}$ strictly decreasing.

G.2 (II) Finite- k Selection $\Rightarrow$ Spectral Shells (Lyapunov Descent)

Linear growth $A_k(t) \sim e^{\sigma(k)t}$ selects an annulus $k \in [k_-, k_+]$. Nonlinear saturation collapses support to shell-quantized radii via

$$\oint k_r(r) dr = 2\pi n, \quad n \in \mathbb{Z}_{\geq 0}.$$

Derivation sketch. The stationary phase $\partial_t \theta = 0$ and bounded $|A|$ imply orbital closure of k_r along rays; Bohr–Sommerfeld quantization yields the shell rule above.

Concrete cavity instantiation (numbers). For a water cavity of length $L = 0.1524$ m, $n_w \simeq 1.33$, the free spectral range (FSR) is $f_{\text{FSR}} = \frac{c \cos \theta_w}{2n_w L} \approx 740$ MHz at $\theta_w = 0$. Identified shell radii in k map to discrete baseband harmonics at $n f_{\text{FSR}}$ after envelope detection.

G.3 (III) Phase Winding $m \in \mathbb{Z} \Rightarrow U(1)$ Fiber

Definition. The phase θ defines a principal $U(1)$ bundle with connection $\mathcal{A}_\mu = \partial_\mu \theta$. Winding $m \in \mathbb{Z}$ over loops implies nontrivial holonomy:

$$\oint \mathcal{A}_\mu dx^\mu = \oint d\theta = 2\pi m, \quad \mathcal{F}_{\mu\nu} = \partial_\mu \partial_\nu \theta - \partial_\nu \partial_\mu \theta = 0 \text{ off zeros of } A.$$

Consequence. $U(1)$ gauge structure emerges from single-valuedness of ψ ; curvature (vortex cores) appears only where $A = 0$ allows branch cuts.

G.4 (IV) Triadic Mode Coupling $\Rightarrow$ $SU(2)$ Structure Constants

Amplitude equations (near onset). For three resonant modes $A_j(t)$ with $\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 = 0$,

$$\dot{A}_i = \mu A_i - g|A_i|^2 A_i - h A_j^* A_k^*, \quad (i, j, k) \text{ cyclic},$$

with real μ, g, h set by projection from the high-order WCT EL equation.

Two-component doublet and commutators. Define a mode doublet $\Phi = (A_1, A_2)^\top$ and real generators $\{\tau_a\}$ acting on Φ by infinitesimal triad-induced rotations:

$$\delta \Phi = i \epsilon^a \tau_a \Phi, \quad [\tau_a, \tau_b] = 2i \epsilon_{abc} \tau_c.$$

Derivation. The cubic vertex $h A_j^* A_k^*$ closes under the triad constraint and supplies an anti-Hermitian coupling on the complex 2D subspace spanned by (A_1, A_2) ; normal-form reduction yields the $\mathfrak{su}(2)$ algebra with structure constants ϵ_{abc} .

Worked 3-mode example (numbers). Take $|\mathbf{k}_j| = k_*$, $\angle(\mathbf{k}_1, \mathbf{k}_2) = 120^\circ$, $\mu = 0.03$, $g = 1$, $h = 0.6$. Starting from $(A_1, A_2, A_3) = (\eta, \eta, \eta)$, the flow selects a locked hexagon. Restrict to (A_1, A_2) and fit infinitesimal rotations from the cubic cross-terms; the commutator numerically recovers $[\tau_a, \tau_b] \approx 2i \epsilon_{abc} \tau_c$ within $< 1\%$ over the attracting set.

G.5 (V) Torsionless Curvature from Feedback $\Rightarrow$ Emergent $g_{\mu\nu}^{\text{eff}}$

Definition (eikonal index). Let the curvature-induced refractive correction be

$$n_{\text{eff}}^2(x) = n_0^2 \left(1 + \chi \frac{\text{Re } W_\psi(x)}{k_0^2} \right), \quad k_0 = \omega/c, \quad \chi > 0.$$

Gordon-type metric. Wavefronts propagate on an effective optical metric

$$ds^2 = c^2 dt^2 - \frac{1}{n_{\text{eff}}^2(x)} d\mathbf{x}^2 = g_{\mu\nu}^{\text{eff}} dx^\mu dx^\nu,$$

with Levi–Civita (torsionless) connection because n_{eff} derives from a scalar potential $\Phi(x) := \chi \text{Re } W_\psi/k_0^2$; i.e., $g_{\mu\nu}^{\text{eff}}$ is conformally flat in the spatial block and $\Gamma_{\mu\nu}^\lambda = \Gamma_{\nu\mu}^\lambda$.

Concrete metric estimate (numbers). In the cavity, take $n_0 = 1.33$, $\chi = 10^{-2}$, $\text{Re } W_\psi/k_0^2 \simeq 10^{-6}$ on a locked shell. Then $n_{\text{eff}} - n_0 \approx \frac{n_0 \chi}{2} 10^{-6} \sim \mathcal{O}(10^{-8})$, giving ppm-level shifts in f_{FSR} .

G.6 (VI) Single Falsifiable Corollary: Shell–Locked Line Shift vs. ΔH

Observable. The baseband ladder $f_n = n f_{\text{FSR}}$ with

$$f_{\text{FSR}} = \frac{c}{2L n_{\text{eff}}(S, H)}, \quad n_{\text{eff}}(S, H) = n_0 + \eta_S S + \zeta_H H^2 + \mathcal{O}(H^4),$$

where $S \in \mathbb{Z}_{\geq 0}$ is the occupied shell index, H an applied static magnetic field, η_S the shell-induced curvature correction, and ζ_H the (weak) magneto-optic susceptibility in the locked state. Then

$$\boxed{\frac{\Delta f_{\text{FSR}}}{f_{\text{FSR}}} = -\frac{\Delta n_{\text{eff}}}{n_{\text{eff}}} \approx -\frac{\eta_S \Delta S + \zeta_H \Delta(H^2)}{n_0}},$$

at fixed L and small corrections. *Prediction.* (i) Stepwise kinks in $f_{\text{FSR}}(t)$ when $S \mapsto S \pm 1$ (lamp/UV toggles), magnitude $\sim |\eta_S|/n_0 = \text{ppm}$; (ii) Quadratic-in- H shift with coefficient ζ_H/n_0 , resolvable at $H \sim 10\text{--}100 \text{ mT}$ if $\zeta_H \gtrsim 10^{-10}\text{--}10^{-9}$.

Numerical target (cavity, $L = 0.1524 \text{ m}$). At $f_{\text{FSR}} \approx 740 \text{ MHz}$, a 5 ppm change yields $\Delta f \approx 3.7 \text{ kHz}$ —well above $\sim 10\text{--}100 \text{ Hz}$ FFT bin widths on long ($\gtrsim 10 \text{ s}$) acquisitions. A quadratic fit of f_{FSR} versus H with shell state held fixed falsifies ζ_H if no curvature-coupled MO response exists.

G.7 (VII) Map to SM/GR (High-Level; SMEFT Deferred)

- $U(1)$ from phase holonomy ($m \in \mathbb{Z}$). Charge $\sim$ winding density; photon-like modes are phase gauge quanta on the $U(1)$ fiber.

- SU(2) from triad closure near a degenerate shell doublet; Pauli algebra emerges as the normal form on the mode doublet under cubic interactions.
- Effective $g_{\mu\nu}^{\text{eff}}$ from curvature feedback $\Rightarrow$ geometric optics in a Gordon metric; weak-field GR phenomenology appears as spatial index modulation from $\text{Re } W_\psi$ (torsionless).

Deferred. Full SMEFT matching (representations, couplings, renormalized vertices) is relegated to the companion paper “*Formal Derivation SM*”.

G.8 One Complete Path (Prove-by-Example)

Object. 2D cavity, $L_x = 0.1524 \text{ m}$, $L_y = 0.1524 \text{ m}$; parameters $(r, a, b, \beta) = (0.02, 1, 1, 1)$; triad hexagon at $k_\star = 1/\sqrt{2}$.

Path.

1. Start at ZW_0 ; small noise excites band $[k_-, k_+]$.
2. Lyapunov descent collapses spectrum to shells: observe integer n via $\oint k_r dr = 2\pi n$.
3. Measure winding m by integrating $\nabla\theta$ around a loop (numerically via unwrapped phase): obtain $m \in \mathbb{Z}$.
4. Identify a near-degenerate doublet (A_1, A_2) and third partner A_3 with $\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 = 0$; fit amplitude equations and extract $\{\tau_a\}$, recovering $[\tau_a, \tau_b] = 2i\epsilon_{abc}\tau_c$.
5. Calibrate n_{eff} from phase delay across L ; fit $n_{\text{eff}} = n_0 + \eta_S S$ holding $H = 0$.
6. Sweep H in $\pm 0\text{--}100 \text{ mT}$ at fixed S and fit $\Delta f_{\text{FSR}}/f_{\text{FSR}}$ vs. H^2 ; infer ζ_H ; compare to the predicted ppm range.

H Density-Weighted Loop–Mass Invariant

Plain definitions

Definition H.1 (Geometry and weight). Let $\Gamma \subset \mathbb{R}^3$ be a closed C^2 space curve of arclength $L_s = \int_\Gamma ds$, with unit tangent $t(s)$, curvature $\kappa(s)$, and torsion $\tau(s)$ (Frenet–Serret). Define the geometric rate

$$\sigma(s) := \sqrt{\kappa(s)^2 + \tau(s)^2} \quad (\text{units: length}^{-1}).$$

Let $w(s) \geq 0$ be a density weight with normalization $\int_\Gamma w(s) ds = 1$. In applications $w \propto \rho = |\psi|^2$.

Definition H.2 (Locked phase rate and effective wavenumber). Let $\theta(s)$ denote the phase accumulated along Γ . The local phase rate is $\theta'(s) = d\theta/ds$. Define the effective wavenumber

$$k_{\text{eff}} := \int_\Gamma w(s) \theta'(s) ds,$$

so that the emergent rest mass/energy are

$$m := \frac{\hbar}{c} k_{\text{eff}}, \quad E_{\text{rest}} := \hbar c k_{\text{eff}}.$$

Symbolic form

Proposition H.3 (Locking principle; density-weighted form). *Consider the locking functional*

$$\mathcal{F}_{\text{lock}}[\theta'] := \int_{\Gamma} w(s) (\theta'(s) - \sigma(s))^2 ds.$$

Its unique minimizer satisfies $\theta'(s) = \sigma(s)$. Consequently,

$$k_{\text{eff}} = \int_{\Gamma} w(s) \sigma(s) ds = \langle \sigma \rangle_w, \quad m = \frac{\hbar}{c} \langle \sigma \rangle_w, \quad E_{\text{rest}} = \hbar c \langle \sigma \rangle_w,$$

where $\langle f \rangle_w := \int_{\Gamma} w(s) f(s) ds$.

Derivation (audit-mode). By convexity, $\mathcal{F}_{\text{lock}}$ is minimized iff its first variation vanishes: $\delta \mathcal{F}_{\text{lock}} = 2 \int_{\Gamma} w(\theta' - \sigma) \delta \theta' ds = 0$ for all $\delta \theta'$, hence $\theta' = \sigma$. Substituting this minimizer in the definition of k_{eff} yields the claim. $\square$

Proposition H.4 (Stability under imperfect lock). *Let $\delta(s) := \theta'(s) - \sigma(s)$. With $\int_{\Gamma} w = 1$,*

$$|k_{\text{eff}} - \langle \sigma \rangle_w| = \left| \int_{\Gamma} w \delta ds \right| \leq \sqrt{\int_{\Gamma} w \delta(s)^2 ds},$$

and therefore

$$|m - m_0| \leq \frac{\hbar}{c} \sqrt{\langle \delta^2 \rangle_w}, \quad m_0 := \frac{\hbar}{c} \langle \sigma \rangle_w.$$

Corollaries (concise)

Corollary H.5 (Planar convex loop; uniform weight). *If Γ is planar convex, $\tau \equiv 0$ and $\int_{\Gamma} \kappa ds = 2\pi n$ with $n \in \mathbb{Z}_{>0}$. If $w \equiv 1/L_s$, then*

$$k_{\text{eff}} = \frac{1}{L_s} \int_{\Gamma} \kappa ds = \frac{2\pi n}{L_s}.$$

Corollary H.6 (Holonomy shifts (U(1) + spin connection)). *Let A be a $U(1)$ gauge potential with flux $\Phi := \oint_{\Gamma} A \cdot dl$, flux quantum $\Phi_0 := h/q$. Let $s \in \frac{1}{2}\mathbb{Z}$ be the spin label and $\Gamma_s := \frac{1}{2\pi} \oint_{\Gamma} \omega$ the (dimensionless) spin-connection index from the Levi-Civita connection ω on the normal bundle of Γ . Then, for perfect geometric lock,*

$$k_{\text{eff}} = \langle \sigma \rangle_w + \frac{2\pi}{L_s} \left(\frac{\Phi}{\Phi_0} + s \Gamma_s \right).$$

Corollary H.7 (Curved-space split). *In a 3-geometry orthogonal to a timelike field u^μ , decompose σ^2 by Gauss–Codazzi:*

$$\sigma(s)^2 = \kappa_g(s)^2 + \tau(s)^2 + K_{\text{ext}}(s), \quad K_{\text{ext}} := n_i n_j {}^{(3)}R_{ikjl} t^k t^l,$$

where κ_g is geodesic curvature in the spatial slice, n a unit normal, and ${}^{(3)}R$ the spatial Riemann tensor. Hence

$$k_{\text{eff}} = \left\langle \sqrt{\kappa_g^2 + \tau^2 + K_{\text{ext}}} \right\rangle_w.$$

Remark H.8 (Finite thickness / medium / dissipation). *Material response and finite core/thickness can be modeled as calibrated factors $\sigma \mapsto \chi_{\text{med}} \sigma$, or as an effective weight $w \mapsto w_{\text{eff}}$ incorporating attenuation; keep calibrations empirical and multiplicative to preserve convexity and the stability bound.*

Worked example (concrete instantiation)

Circle. For a circle of radius R : $\kappa = 1/R$, $\tau = 0 \Rightarrow \sigma = 1/R$. With any w constant along Γ , $\langle \sigma \rangle_w = 1/R$, so

$$m = \frac{\hbar}{cR}, \quad E_{\text{rest}} = \frac{\hbar c}{R}.$$

Numerics: $R = 1 \mu\text{m} = 10^3 \text{ nm} \Rightarrow E_{\text{rest}} \approx \frac{197.326 \text{ eV}\cdot\text{nm}}{10^3 \text{ nm}} \approx 0.197 \text{ eV}$.

Helix. For a circular helix of radius r and pitch p (advance per 2π turn), the curvature–torsion invariants are constant:

$$\kappa = \frac{r}{r^2 + (p/2\pi)^2}, \quad \tau = \frac{(p/2\pi)}{r^2 + (p/2\pi)^2} \Rightarrow \sigma = \sqrt{\kappa^2 + \tau^2} = \frac{1}{\sqrt{r^2 + (p/2\pi)^2}}.$$

With any constant w , $\langle \sigma \rangle_w = \sigma$, so

$$E_{\text{rest}} = \hbar c \sigma = \frac{\hbar c}{\sqrt{r^2 + (p/2\pi)^2}}.$$

Numerics: $r = 100 \text{ nm}$, $p = 200 \text{ nm} \Rightarrow \sqrt{r^2 + (p/2\pi)^2} \approx 105 \text{ nm}$, so $E_{\text{rest}} \approx 197.326/105 \text{ eV} \approx 1.88 \text{ eV}$.

Appendix pointers (for full paper)

- **Holonomy:** derive Cor. H.6 via Stokes ($\Phi = \int B \cdot dS$) and the spin connection line integral on the normal bundle.
- **Curved-space split:** derive Cor. H.7 from Gauss–Codazzi equations for curves on embedded surfaces in a 3-slice.
- **Stability:** prove Prop. H.4 using Cauchy–Schwarz with the weight w .
- **Calibration:** tabulate χ_{med} for media/thickness; specify how w_{eff} is constructed from measured $\rho = |\psi|^2$.

I Computational Review Note

Several numerical and logical assessments in this paper were assisted by ChatGPT, an AI language model developed by OpenAI. The model was used to check order-of-magnitude estimates, structural coherence, and consistency with existing scientific literature. All factual claims remain the responsibility of the author.